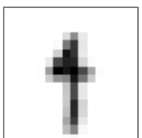
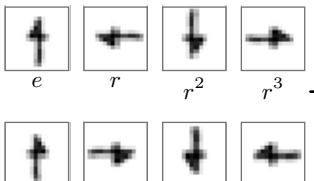
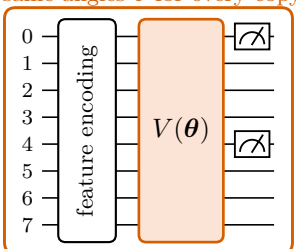


image x

copies $R_{\mathcal{X}}(g)x$, $g \in D_4$



NonEquiv circuit,
same angles $\boldsymbol{\theta}$ for every copy



mean of the eight outputs

$$f_{\mathrm{tw}}(x) = \frac{1}{8} \sum_{g \in D_4} f_{\mathrm{neq}}\big(R_{\mathcal{X}}(g)x\big)$$

invariant: $f_{\mathrm{tw}}\big(R_{\mathcal{X}}(h)x\big) = f_{\mathrm{tw}}(x)$

eight circuit evaluations per image